

E. Ein es-Sultan

time limit per test: 1 second
memory limit per test: 1024 megabytes

Ein es-Sultan is a natural spring in Jericho, at the site of ancient Jericho, which has been identified with the tell (archaeological mound) known as Tell es-Sultan. It is known by Jews and Christians as the Spring of Elisha/Elisha's Spring, based on a biblical story about Prophet Elisha. In 2010, the spring saw the end of a year-long rehabilitation programme. By 2000 a protective building which helps avoid contamination of the water had been erected over the spring, and by 2010 the rehabilitation included the old facilities, the preservation of archaeological remains, and landscaping works at the site for tourism purposes.

You are standing at the fourth vertex of a complete graph with four vertices. You want to count the number of paths that have a length of at most n and end at the starting vertex.

Complete graph: is a graph that every pair of node have an edge between them.

As the result can be very large, you should print the answer modulo $10^9 + 7$

Input

The only line contains one integer n ($1 \leq n \leq 10^{16}$) — the maximum length of a path.

Output

Print integer c — the number of paths.

Examples

input	Copy
4	
output	Copy
30	
input	Copy
14	
output	Copy
1793613	
input	Copy
1	
output	Copy
0	

ACPC Scientific Committee Archive

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About Group

https://www.youtube.com/@MoamenAhmed98/play

Group website

Group Contests

• TCPC 2025

• ACPC Girls 2025

• ACPC Girls 2020

• HPV-CPC 2021

• Tunisian Collegiate Programming Contest (2017)

• Azhar Collegiate Programming Contest 2019

• The 2018 Menofia University Collegiate Programming Contest

• ICPC International Collegiate Programming Contest, Assiut Collegiate Programming Contest (2019)

• International Collegiate Programming Contest, GUC Collegiate Programming Contest (2019)

• International Collegiate Programming Contest, Aleppo University Collegiate Programming Contest(2018)

• The 2025 ICPC Egyptian Collegiate Programming Contest for Teens(Online Qualification)

• ECPC Qualifications 2021 - Day 4

• ACPC Kickoff 2025

• International Collegiate Programming Contest, Al-Baath, Tishreen, Aleppo and Damascus Collegiate Programming Contest (2021)

• The 2020 Egyptian Collegiate Programming Qualification Contest Day 2

• OCPC 2025 (Qualification)

• OCPC 2025

• ACPC 2024 (Mirror)

• LCPC 2024 (Mirror)

• JCPC 2024 (Mirror)

• EthCPC (Mirror)

• PCPC 2024

• LyCPC 2024

• AOCPC 2024

• AOCPC 2023

• QCPC 2024 (Mirror)

• TCPC 2024 (Mirror)

• AICPC 2023

• SCPC 2024 (Mirror)

• SCPC 2024 For Teens (Mirror)

• ECPC 2024 (Mirror)